NOTE THIS IS NOT SEVCON PIN COMPATIBLE. DO NOT CONNEC THIS TO SEVCON
IT WILL BLOW UP

J6
776231-1

BMS CAN1 GND -> CAN1_GND 1
BMS CAN1 LOW -> CAN1_LOW 2
BMS CAN1 HIGH -> CAN1_HIGH 3
BMS CAN1 +5V -> CAN1_+5V 4
5
REDUNDANT THROTTLE B INPUT -> THROTTLE2_IN_A 6
REDUNDANT THROTTLE A INPUT -> THROTTLE1_IN_A 7
THROTTLE GROUND -> GND_A 8
THROTTLE +5V -> +5V_A 9
CONNECT SWITCH BETWEEN +12V_A AND THIS PIN -> I_+12V_KEYSWITCH_A 10
BLANK TO ISOLATE TRACTION BATTERY VOLTAGE -> 11
TRACTION BATT+ (max 160v) -> I_+VIN_Z 12
CAN1 TERMINATION JUMPER -> CAN1_JUMPER_A 13
CAN1 TERMINATION JUMPER -> CAN1_JUMPER_B 14
CAN2 TERMINATION JUMPER -> CAN2_JUMPER_A 15
CAN2 TERMINATION JUMPER -> CAN2_JUMPER_B 16
AUX 12V OUT, TO DISPLAY, KEYSWITCH -> +12V_SWITCHED_A 17
USER DIGITAL INPUT 1 (see ctrl board for v/current limits) -> DIN_1_W 18
DIN GND -> GND_W 19
THROTTLE HOME POSITION SWITCH (REF'D TO THROTTLE +5V) -> +5V_IDLE_SWITCH_A 20

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

21 GND_A <- GND REFERENCE FOR MOTOR TMP, ENCODER
22
23 O_PRECHARGE_DC_LINK_+_Z <- CONNECT TO DC LINK+, ALLOWS INVERTER TO PRECHARGE DC LINK
24 CAN2_GND <- DASH, CHARGER, ABS CANBUS GND
25 CAN2_LOW <- DASH, CHARGER, ABS CANBUS LOW
26 CAN2_HIGH <- DASH, CHARGER, ABS CANBUS HIGH
27 CAN2_+5V <- DASH, CHARGER, ABS CANBUS +5V
28 MOTOR_TEMP_IN_A <- MOTOR THERMISTOR (TIE OTHER END OF THERMISTOR TO GND_A)
29 HALL_W_IN_A <- MOTOR HALL W (IF USED, IF SIN/COS LEAVE N/C)
30 HALL_V_ENCODER_COS_IN_A <- MOTOR HALL V OR ENCODER COSINE (DEPENDING ON SENSOR TYPE)
31 HALL_U_ENCODER_SIN_IN_A <- MOTOR HALL U OR ENCODER SIN (DEPENDING ON SENSOR TYPE)
32 +5V_A <- MOTOR ENCODER/TMP SENSOR +5V
33
34 O_+VIN_SWITCHED_Z <- OUTPUT, TIE TO SEVCON DC/DC ENABLE PIN
35 GND_Z <- TRACTION BATTERY MINUS TERMINAL
MH1 MH2 MH3 MH4

21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 MH1 MH2 MH3 MH4

Connector to IO board
J7
TSW-108-18-F-D

HALL_U_ENCODER_SIN_IN_A 1 — 16 DOUT4_A
+5V_A 2 — 15 DOUT3_A
HALL_V_ENCODER_COS_IN_A 3 — 14 DOUT2_A
GND_A 4 — 13 DOUT1_A
HALL_W_IN_A 5 — 12 GND_A
GND_A 6 — 11 THROTTLE2_IN_A
MOTOR_TEMP_IN_A 7 — 10 GND_A
GND_A 8 — 9 THROTTLE1_IN_A

SINGLE WIRE DEBUG / PROGRAMMING
USB INTERFACE TO TOP IO BOARD
+12V TO CTRL BOARD FROM IO BOARD
J14
TSW-110-18-T-D

GND_A 1 — 20 +12V_IN_A
MP_SWO 2 — 19 GND_A
MP_BOOTSEL 3 — 18 CP_BOOTSEL
MP_SWCLK 4 — 17 CP_SWCLK
MP_SWDIO 5 — 16 CP_SWDIO
MP_RESET 6 — 15 CP_RESET
GND_A 7 — 14 +3.3V_A
USB_DP 8 — 13 USB_DM
MD_FIBER_TX 9 — 12 MD_FIBER_SYNC
MD_FIBER_RX 10 — 11 FAULT

I_+5V_IDLE_SWITCH_A — DIN_1_W
I_+12V_KEYSWITCH_A — DIN_2_W
GND_A — GND_W

USB Debugging port
12Mbit/s - not powered
inverter must have external power

J1
UJ2-B-HR-G-V

GND_A
MH2
USB_DM 1 — VBUS
2 — D-
GND 4 — GND_A
D+ 3 — USB_DP
MH1
GND_A

H1 971250581
H2 971250581
H3 971250581
H4 971250581
H5 971250581
H6 971250581
H7 971250581
H8 971250581

C47 UCM1H101MCL1GS GND_A 2 — 1 +12V_A
C48 UCM1H101MCL1GS GND_A 2 — 1 +12V_A
C49 UCM1H101MCL1GS GND_A 2 — 1 +12V_A
C42 UCM1H101MCL1GS GND_A 2 — 1 +12V_A
C43 UCM1H101MCL1GS GND_A 2 — 1 +12V_A

onboard isolated 40-160v -> 12v 20W power supply

INPUT POLARITY PROTECTION
F1
MF-RHT200_32-2
D3
S3N
D4
S3N
I_+VIN_Z
VIN_FUSED_Z
GND_Z

PS3
EC7BW-110S12
+12V_IN_A +12V_A
C189 100nF
GND_A +12V_A
REMOTE_ON/OFF 6 — TRIM 4
-VIN 2 — -VOUT 5 — GND_A
+VIN 1 — +VOUT 3 — +12V_A

Connector to IO board
J12
TSW-108-18-F-D

DIN_1_W 1 — 16 GND_W
GND_W 2 — 15 DIN_8_W
DIN_2_W 3 — 14 GND_W
GND_W 4 — 13 DIN_7_W
DIN_3_W 5 — 12 GND_W
GND_W 6 — 11 DIN_6_W
DIN_4_W 7 — 10 GND_W
GND_W 8 — 9 DIN_5_W

EXPANSION MODULE CONNECTOR
(FOR FIBER IO MULTIDRIVE MODULE, ETC)
ALLOWS MULTI-DRIVE MANY PHASE (>3) OR LOAD SHARING
J2
SSW-104-01-L-D

MD_FIBER_TX 1 — 8 MD_FIBER_SYNC
MD_FIBER_RX 2 — 7 FAULT
+5V_A 3 — 6 GND_A
+12V_A 4 — 5 GND_A

+12V external switch
enables/disables aux 12v output.
Note, when system is running, this output must not be on
due to power constraints with onboard regulator
intended use case is to run aux systems while
traction power is not applied when trpwr is on,
offboard main power system should power loads instead.

+12V_A
3
U6
BTS462TATMA1
DOUT4_A 2 — IN
Vbb
GND
OUT 5 — +12V_SWITCHED_A
GND_A 1

Connector to IO board
J13
TSW-103-18-G-D

CAN2_JUMPER_A 1 — 6 CAN2_HIGH
CAN2_JUMPER_B 2 — 5 CAN2_LOW
CAN2_+5V 3 — 4 CAN2_GND

J10
TSW-103-18-G-D

CAN1_JUMPER_A 1 — 6 CAN1_HIGH
CAN1_JUMPER_B 2 — 5 CAN1_LOW
CAN1_+5V 3 — 4 CAN1_GND

IC10
DRDNB21D-7

DOUT1_A 4 — B2 — C2/A2 3
GND_A 5 — K1/K2 — E1/E2 2 — +5V_A
DOUT2_A 6 — B1 — C1/A1 1

C188 100nF
GND_A — +5V_A

PRECHARGE CIRCUIT
K1
AWHSH105D00G

VIN_FUSED_Z 2
NO 1
NC 3
COIL_1 — COIL_2 4 — +5V_A
COM 5 — O_PRECHARGE_DC_LINK_+_Z

NOTE TO USER:
THEY MUST CONNECT EXTERNAL PRECHARGE RESISTOR
SUCH THAT <2A FLOWS

K2
AWHSH105D00G
SEVCON DC/DC ENABLE CIRCUIT

VIN_FUSED_Z 2
NO 1
NC 3
COIL_1 — COIL_2 4 — +5V_A
COM 5 — O_+VIN_SWITCHED_Z

HW-C2-PCB-IO
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: IOBoard.kicad_sch

Title: Open Inverter Platform - IO Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 1/1